

# Cloud Computing Lab 1

## Setting Up a Virtual Machine with Oracle VirtualBox

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**Date** 22 August 2026

**Hypervisor** Oracle VirtualBox 7.2.16

**Guest OS** Ubuntu 25.04 (Plucky Puffin), 64-bit

### 1.OBJECTIVE

The objective of this lab is to install an open-source hypervisor, create a virtual machine, configure its CPU, RAM, storage and network settings, attach a guest operating system ISO image, boot the VM, complete the guest OS installation, and verify that the VM runs successfully.

### 2. Tools Used

- Hypervisor: Oracle VirtualBox 7.2.16
- Guest OS image: ubuntu-26.04-desktop-amd64.iso
- Host OS: Windows
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### 3. Step 1 — Download and Install VirtualBox

Oracle VirtualBox was downloaded from the official website (virtualbox.org). The platform package for Windows hosts was selected, along with the VirtualBox Extension Pack, which was accepted under the PUEL license.

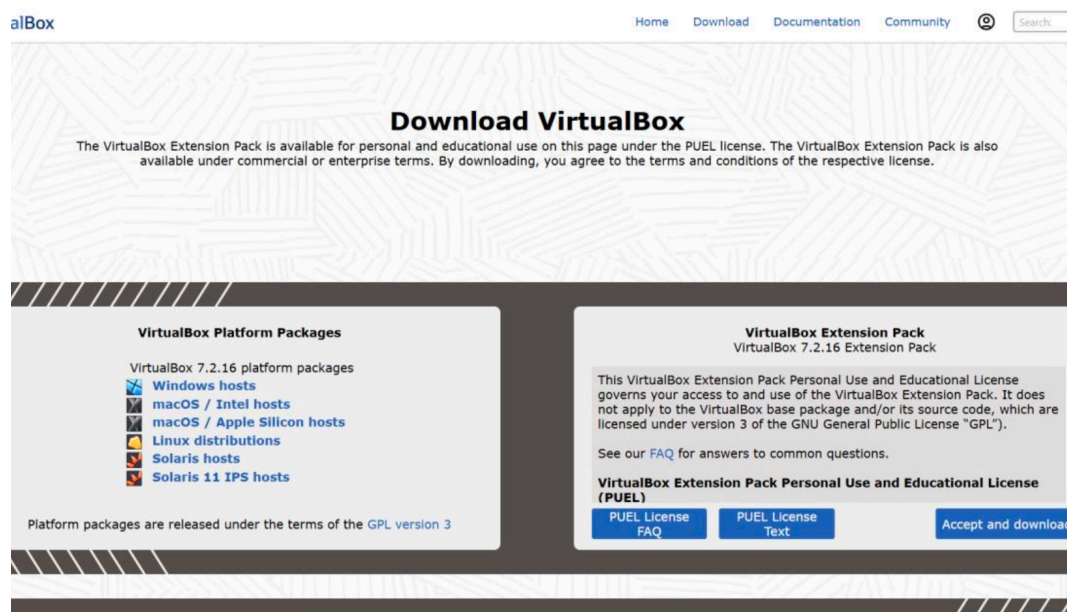


Figure 1: VirtualBox download page showing platform packages and the Extension Pack.

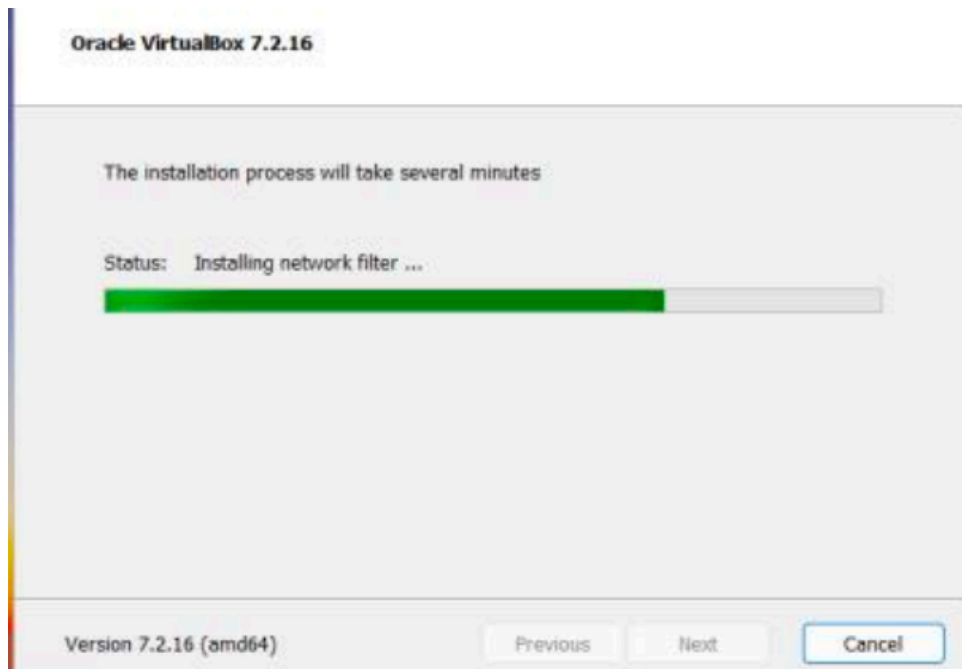


Figure 2: Oracle VirtualBox 7.2.16 installer in progress.

#### 4. Step 2 — Launch VirtualBox Manager

After installation, Oracle VirtualBox Manager was launched. On first run, the Experience Mode selection screen appeared; Expert Mode was selected to access all configuration options while creating the VM.

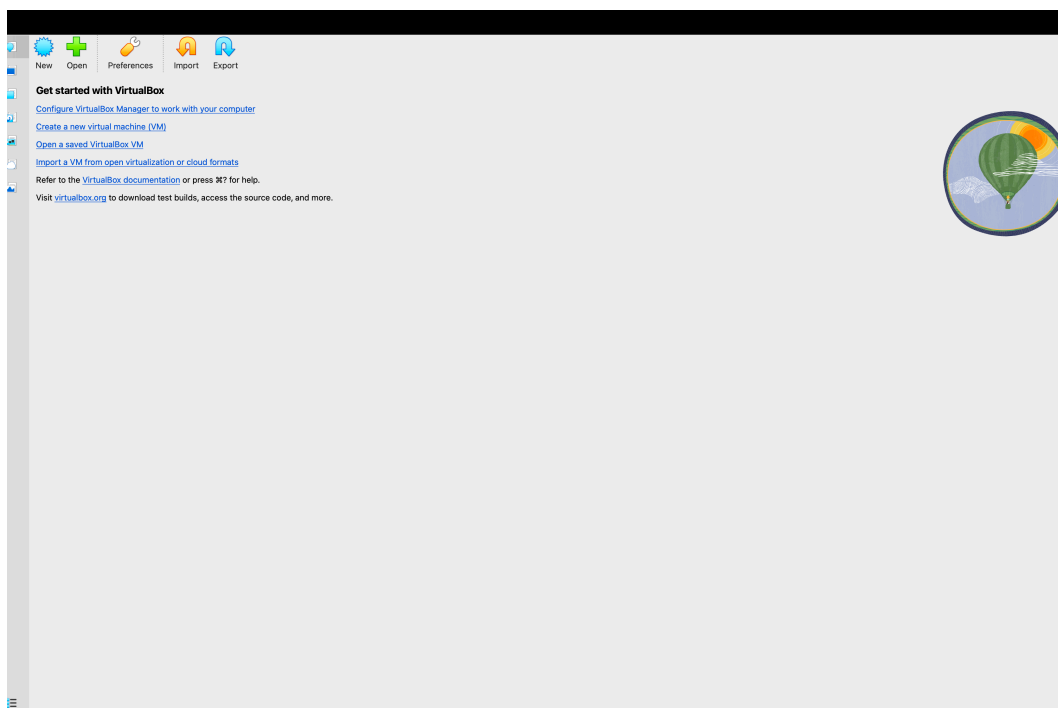


Figure 3: Oracle VirtualBox Manager home screen with Experience Mode selection.

## 5. Step 3 — Create a New Virtual Machine

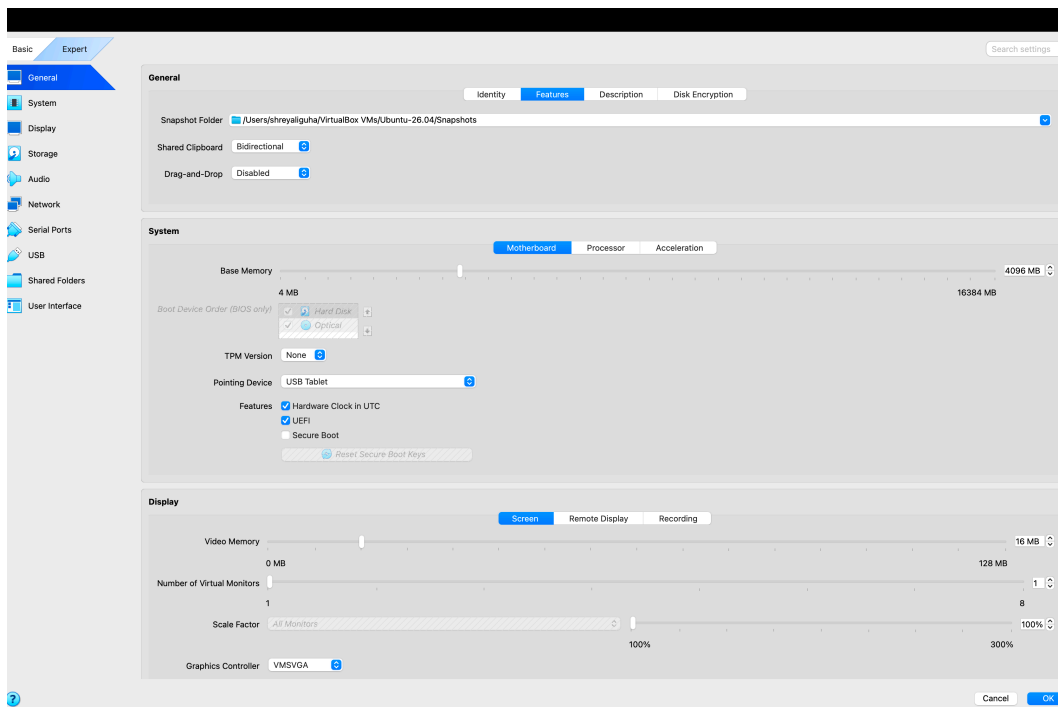


Figure 4: New Virtual Machine wizard — VM name, folder, and ISO image attached.

## 6. Step 4 — Configure CPU, RAM, and Storage

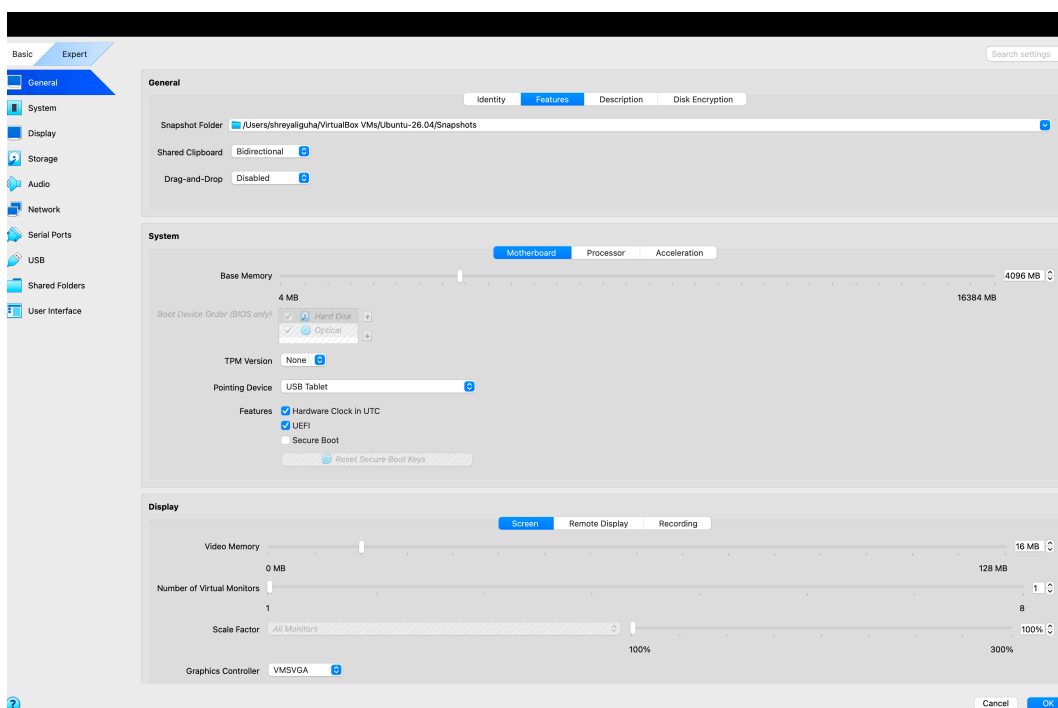


Figure 5: Specifying virtual hardware — base memory, CPU count, and disk size.

## 7. Step 5 — Review Configuration Summary

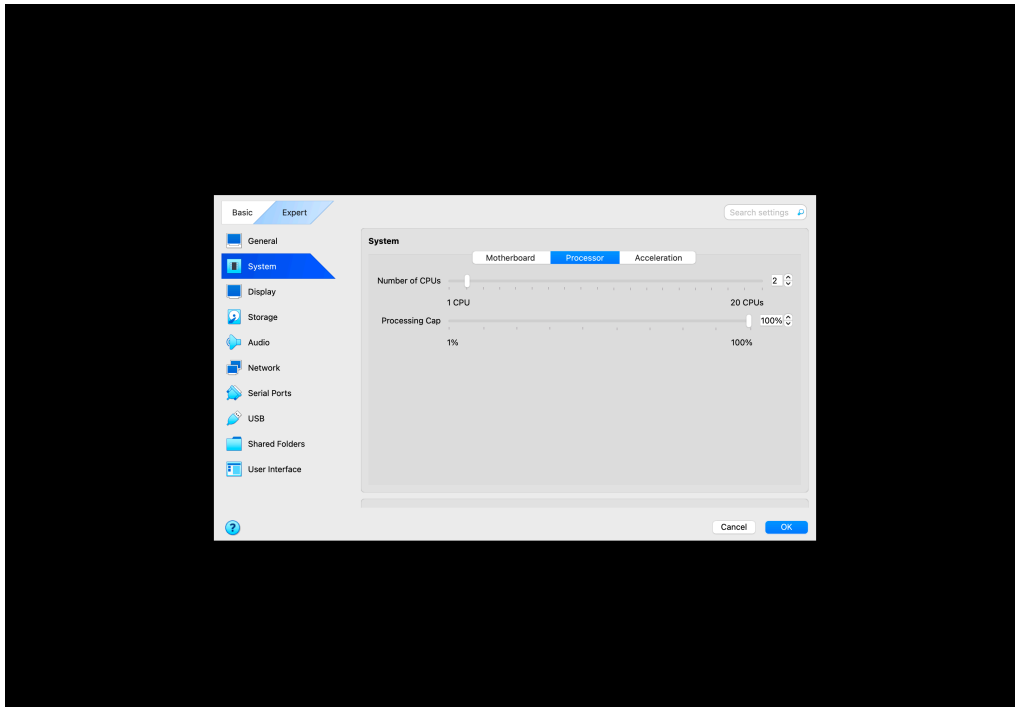


Figure 6: Summary screen showing the final VM configuration.

## 8. Step 6 — Verify Network and General Settings

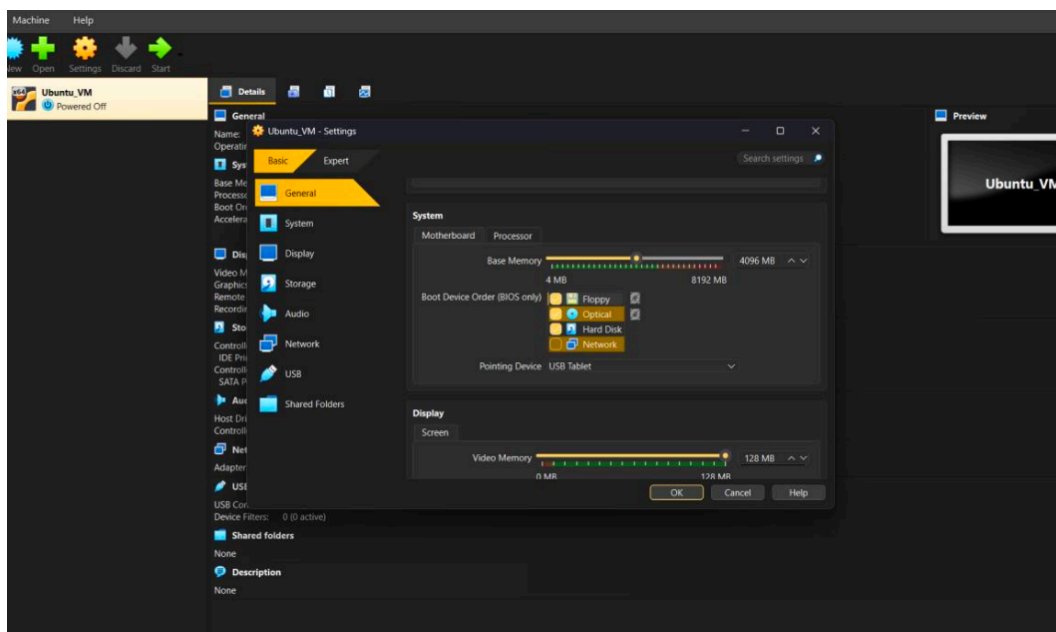


Figure 7: VM Settings — General/System configuration review

## 9. Step 7 — Boot the VM and Install the Guest OS

## 10. Step 8 — Install Build Tools and Kernel Headers

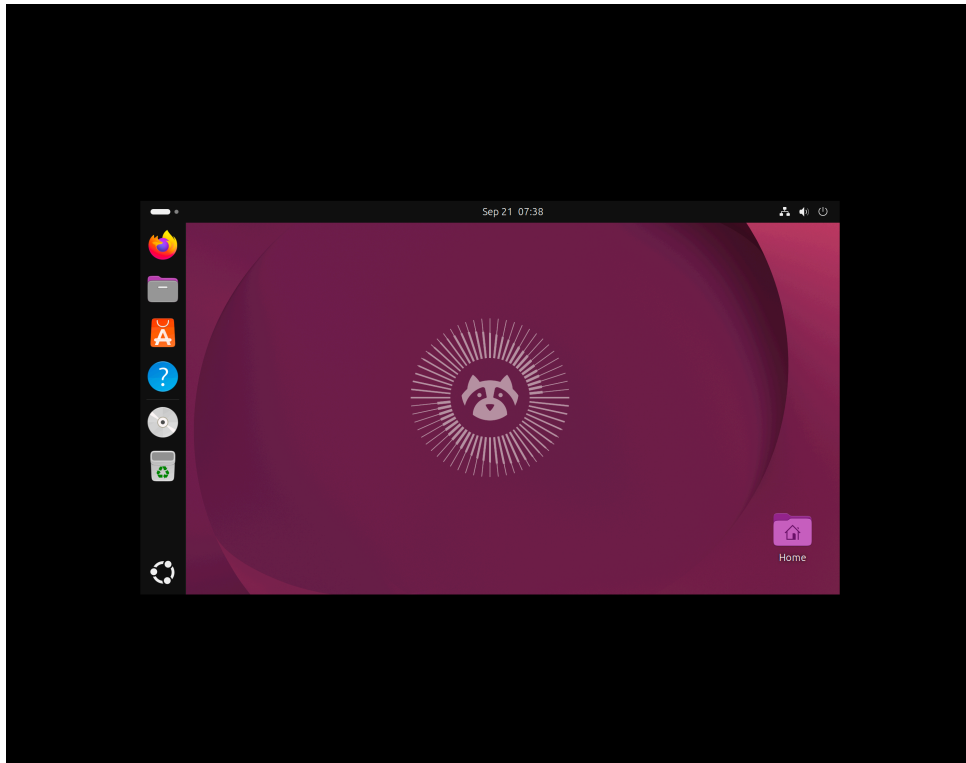


Figure 8: Ubuntu desktop running successfully inside the VirtualBox VM.

Once the VM was running, a terminal was opened inside the Ubuntu guest and the package index was updated. The build-essential package and the Linux kernel headers matching the running kernel were then installed using the following commands:

```
sudo apt update  
sudo apt install -y build-essential linux-headers-$(uname -r)
```

Additions.

### 11. Verification

- VirtualBox 7.2.16 installed successfully on the host machine.
- A new VM (Ubuntu\_VM) was created with 4096 MB RAM, 2 CPUs, and a 25 GB virtual disk.
- The Ubuntu ISO image was attached and the guest OS was installed.
- The VM booted successfully into the Ubuntu desktop environment.
- Build tools and kernel headers were installed inside the guest OS.

### 12. Conclusion

This lab demonstrated the complete workflow of setting up virtualization on a local machine: installing a hypervisor (Oracle